

Project Proposal: Fraudulent Review Detection in Apartment Reviews in Orem and Provo UT

Problem Statement:

Online reviews can provide good insight on the quality of a product or service. They can help give a potential customer a more accurate view of the experience of a real person with the product or experience. However, in Orem and Provo Utah, students seeking housing are running into a problem. There have been reports that the reviews that are on google maps or other review websites may tend to be innaccurate or skewed, giving students a harder time understanding what the real experience is living at a certain apartment. The primary suspicion is that the apartment complexes are providing themselves fake reviews or having their employees write positive reviews to boost the perception of their apartment complex in a positive light.

Solution

A model that can accurately take information about a review or a distribution of reviews and provide information about whether or not the review is deceptive or not.

Plan

Milestone	Work Carried Out	Outcome/Deliverable	Completion Date
Milestone 1	- Data Collection - Scraping google web reviews - Collecting data from peers and locals - Collect training data from other studies - Appropriately label training data	- MongoDB database - Contains records of reviews, rankings, etc...	- Scraping Google Reviews: Week 1 - Collecting Traing Data: Week 2 - Collecting data from peers: Week 5
Milestone 2	- Exploratory Data Analysis - Explore different types of reviews - Explor what kind of data is available - Compare scraped reviews to local reviews	- Statistical report on reviews - Report on distribution of ratings - Report on individual reports - Report on elementary sentiment analysis - Report on potential features to extract	Reports on week 5, 6, 7
Milestone 3	- Modeling and experimentation - Apply other tested models on my data - Experiment with newer models on my data	- Results from model experiments - Creating actual model - Deploying actual model into an app (if time)	Weeks 8, 9, 10